

DHRITI MITTAL

[LinkedIn](#) • [GitHub](#) • dhritimittal191205@gmail.com • +91 9975717082

Full Stack Engineer with hands-on experience in software testing, automation, and applied AI/ML. I don't trust AI outputs by default — I engineer the systems that catch what they get wrong. Built production web systems, multi-agent RAG applications, distributed ML infrastructure, and applied deep-learning models.

EDUCATION

B.Tech in Computer Science and Engineering (Core) | Vellore Institute of Technology 2023 – Present
CGPA: 8.29

Class 12th (CBSE) | Viraj Shri Ram Centennial School, Maharashtra 2023
Percentage: 87.2%

Class 10th (CBSE) | Viraj Shri Ram Centennial School, Maharashtra 2021
Percentage: 99.6%

SKILLS & CERTIFICATIONS

Programming: Python, Go, C++, TypeScript/JavaScript, SQL

AI/ML: PyTorch, TensorFlow, Deep Learning, Computer Vision, RAG, Prompt Engineering, Multi-Agent Systems

Full Stack: React, Next.js, FastAPI, REST APIs, Postman, gRPC, Git/GitHub, CI/CD

Certifications: NPTEL - Introduction to Machine Learning; NPTEL - Marketing Analytics; Google/Coursera - Bits and Bytes of Computer Networking; Adobe - UI/UX; ServiceNow - Virtual Internship.

EXPERIENCE & LEADERSHIP

Technical Summer Intern | LearningShala June - July 2026
Working directly with the founder, owning product strategy, UI/UX, frontend architecture, and API integration.

Project – College Discovery & Counselling Platform [\(Live Link ↗\)](#)

- Analyzed and redesigned course discovery, admissions, and counselling user journeys in React/Next.js/Tailwind, iterating directly with the founder to reduce user friction and improve conversion.
- Replaced hardcoded styling with a semantic design-token system in Tailwind v4, enabling scalable multi-university theming and reducing repeated page-level styling work.
- Migrated 250+ course pages from static data to an API-driven CMS pipeline, restructuring nested data and resolving Vercel build/configuration failures to restore reliable production delivery.
- Tested and validated the REST API endpoints powering the CMS pipeline using Postman, catching integration issues before they reached production.

Lead, Operations Department | VIT Bhopal University — Android Club April 2024 - April 2026

- Coordinated 80+ cross-functional members across 25+ technical events serving 1,000+ participants, managing schedules, logistics, stakeholder communication, and budgets.
- Streamlined event planning and coordination workflows, reducing planning time by 20% while delivering events on schedule.

Co-Lead, Technical Department | VIT Bhopal University — Matrix Club January - September 2025

- Led 5 UI/UX and Figma workshops for 25+ members; 80% reported increased confidence in design tooling in post-workshop surveys.

PROJECTS

RepoSearch AI: Autonomous Codebase Analyst | LangChain, LangGraph, FastAPI, ChromaDB, RAG [\(Link ↗\)](#)

- Built an autonomous codebase-analysis system that routes complex developer queries across 6 task-specialized agents for bug hunting, refactoring, and repository analysis.
- Designed a hybrid RAG pipeline combining BM25, MMR, cross-encoder reranking, and language-aware code chunking to improve retrieval quality across multi-language codebases.
- Built a FastAPI/Next.js application with real-time SSE streaming and privacy-preserving offline analysis via Ollama.

MeshLLM: Distributed Inference Network | Go, Python, gRPC, Tailscale [\(Link ↗\)](#)

- Built a distributed inference network for executing Llama 3 across fragmented remote GPUs using pipeline parallelism and a Go-based orchestration server.
- Implemented node heartbeats and API-key issuance over gRPC/HTTP and established peer-to-peer model-weight and activation transfer through a Tailscale mesh.

Physics-Informed Projectile Prediction Model | PyTorch, Computer Vision, PINNs

- Architecting a hybrid CV/PINN system to forecast projectile motion from video while enforcing physical constraints through a custom physics-guided loss.
- Designing a CNN backbone to extract trajectory/state variables from noisy or occluded video frames; targeting a 90% reduction in prediction drift versus conventional CV baselines.